GOKUL NAATH M

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EDUCATION

Chennai Institute of Technology Chennai, TN
Bachelor of Engineering in Electrical and Electronics (CGPA-8.93) *Sep. 2023 – May 2027*

Velammal Matriculation Higher Secondary School Chennai, TN
HSC 2023: 89 /100 and SSLC 2021 *Jun. 2021 – Apr. 2023*

INTERNSHIPS

Flextronics Technologies pvt ltd (May 2024)

- At Flex, I had the opportunity to work on AI and ML-based projects, where I applied machine learning algorithms to solve real-world problems. I collaborated with cross-functional teams to develop data-driven solutions, leveraging techniques like data analysis, model training, and optimization. This experience helped me sharpen my analytical and technical skills, as well as deepen my understanding of AI/ML applications in the manufacturing and technology sectors.

Research Internship at Chennai Institute of technology (Nov 2024)

- Review paper 1 = Smart Diagnosis using AI and ML Driven Fault Detection in Integrated Circuit.
- Research paper 1= WIND TRACK- Generation of Renewable energy from train induced wind flows

Sakthi Transformers, Thirumudivakkam (May 2025)

- Interned at Sakthi Transformers, gaining hands-on exposure to transformer manufacturing and testing, including core construction, coil winding, tank fabrication, oil filtration, and routine electrical tests using modern tools. Learned testing procedures like insulation resistance, TTR, and oil BDV using instruments such as megger and 3 -phase power analyzer. Gained insight into quality assurance, industrial safety practices, and standard compliance in transformer production.

PROJECTS & ACHIEVEMENTS

Wind Track - Generation of Renewable energy from train induced wind flows(SIH 2024 ,Semi Final)

- Developed a sustainable energy system using Savonius vertical-axis wind turbines placed between railway tracks to harness wind generated by moving trains. Designed and tested a low-cost setup capable of producing up to 15 kWh per train pass, with applications in powering railway infrastructure.

Piezo Electric Floor Tile (Electrothon 2024 ,Finals)

- Piezo electric flooring uses crystal that generates electricity when it is subjected to pressure or vibration.it could revolutionize urban infrastructure by providing renewable energy from human activities. The main output is to create the world where every step can generate electricity.

IoT-Based Real-Time Oil Flow Monitoring System (Mploy Company, 2025)

- Developed an IoT-enabled oil monitoring system using ESP32, ultrasonic and flow sensors to track oil levels and flow rates in real-time. Integrated Google Sheets for live data logging and visualization, enabling remote access and analysis. Implemented motor control automation based on tank level thresholds, and built a companion web dashboard with pie chart analytics for daily oil usage tracking.

Smart Monitoring System for Water & Power Consumption (SDG Hackathon 2025, Finals)

- Developed an IoT-based monitoring system using ESP32 to track water flow, tank level, and electrical parameters. Logged real-time data to Google Sheets and applied machine learning (Linear Regression) for hourly prediction of water and power usage, enabling smart resource management for residential or commercial applications.

Smart Solar Panel Cleaning & Efficiency Enhancement System

- Designed and developed an automated maintenance solution for solar panels using a microcontroller-based smart control system, motor-driven rotating brush, and controlled water spray mechanism to remove dust, dirt, and environmental impurities. The system helps to improve photovoltaic efficiency, maintain consistent power output, reduce manual cleaning effort, minimize maintenance cost, and support the effective utilization of renewable energy resources.

Python Game Development

- UFO ATTACK | DESTROYONG ASTERIODS | COWS AND BULLS (https://github.com/Gokulnaxth)

COURSES AND CERTIFICATIONS

- INTRODUCTION TO PYTHON
- PYTHON GAME DEVELOPMENT
- HTML TUTORIAL
- ELECTRONICS MASTERY
- INCREDIBLE ELECTRONICS
- INTRODUCTIONTO EV
- PROMPT ENGINEERING

TECHNICAL SKILLS

Software Languages: Python, C, Java, My SQL, Thonny Software
Tools: Microsoft Excel | Blender | LT SPICE | FRONIUS